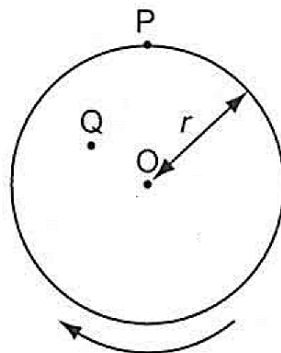


- 26** A copper disc of radius r rotates about its centre O at a constant speed. It is placed in a uniform magnetic field perpendicular to its surface. P is a point on the rim of the disc, while Q is a point at distance $\frac{r}{2}$ from O .



A steady electromotive force (e.m.f.) E is generated between points O and P .

What is the e.m.f. generated between points P and Q ?

A zero

B $\frac{1}{4} E$

C $\frac{1}{2} E$

D $\frac{3}{4} E$

Answer: D

Between O and P , $e.m.f. = B(\pi r^2)f = E$

Between O and Q , $e.m.f. = B(\pi(r/2)^2)f = \frac{1}{4} E$

Between P and Q , $E - \frac{1}{4} E = \frac{3}{4} E$

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A half wave rectified sinusoidal alternating current flows through a light bulb. The graph shows